



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2023/2024

SECV2113 - HUMAN COMPUTER INTERACTION

SECTION 1

ASSIGNMENT 1

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a) Purpose of the technology

Our group has chosen “Ecosia”, a persuasive technology that promotes SDG 15: Life on Land, one of the 17 Sustainable Development Goals. Ecosia is a search engine that differs from other popular search engines such as Google and Bing because of its unique purpose and objective. Unlike other search engines, which mostly rely on advertising to generate revenue, Ecosia allocates its ad revenue to fund global tree-planting projects and other environmental conservation activities.

The purpose of Ecosia is to use the power of internet searches to benefit the environment. Every time a user makes a search on Ecosia, the search engine generates money via advertisements that appear alongside the search results. The majority of this money is subsequently used by Ecosia to fund tree-planting initiatives around the world, which help to prevent climate change, preserve biodiversity and reforest areas.

Ecosia's mission is based on sustainability and environmental stewardship. By utilizing Ecosia as their preferred search engine, users actively support forestry programs, contribute to carbon sequestration, and encourage sustainable land management methods. The main objective of Ecosia is to make the world more sustainable and greener by leveraging the collective power of internet users' daily actions.

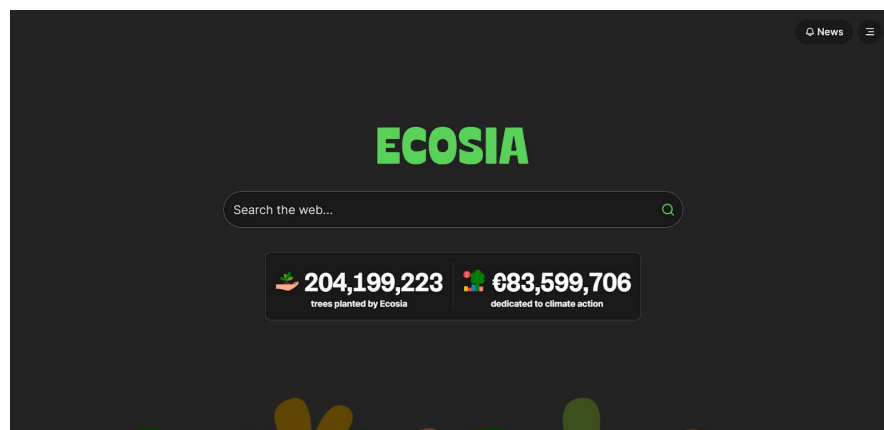
b) 2 specific task/feature

Feature 1: Ecosia's Tree-Planting Progress Tracker

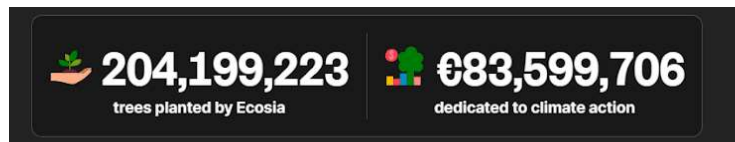
Target Behavior: Checking the progress of tree-planting efforts on Ecosia.

The website for Ecosia effectively combines triggers, motivation, and ability to encourage users to support the practice of using Ecosia for their searches. The counter encourages users to maintain their using Ecosia to encourage tree planting initiatives by acting as a visual reminder that highlights the benefits of utilising the platform.

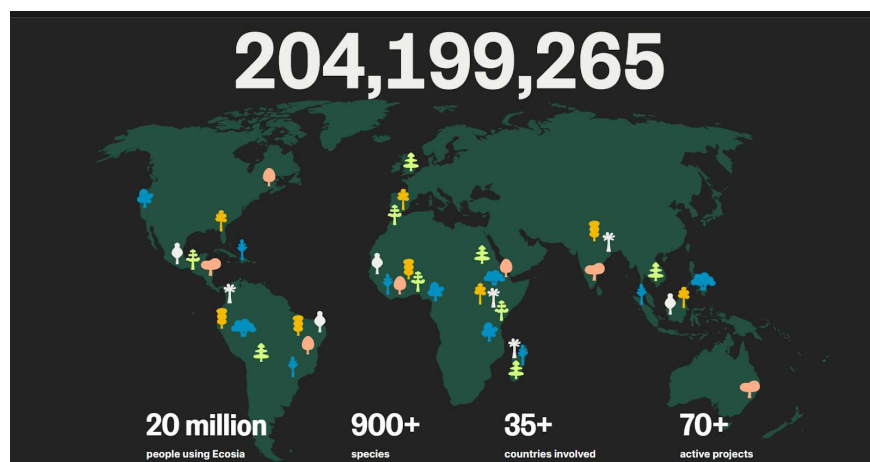
Ability - Ecosia's progress tracker and user-friendly search engine make it easy for users to track tree-planting progress and switch to using Ecosia for their searches. The homepage features an obvious progress tracker that makes it straightforward to track the number of trees planted without requiring complicated actions. Its combination with an identifiable UI similar to other well-known search engines reduces the entry barrier and promotes regular usage.



Motivation - Users are motivated by the measurable and visible influence they can make on reforestation via Ecosia. For example, a user may experience a feeling of success and be inspired to keep using Ecosia to expand their effect if they find that their searches have planted 1,000 trees.



Trigger - The tree planting counter on Ecosia's website serves as a visual trigger by displaying the quantity of trees planted, it continuously reminds the user of the environmental effect of their searches. Users are motivated to keep using Ecosia by this sense of advancement and success since they are pleased with the contribution they have made. For instance, every time a user opens a new tab, they can see the increase of the amount of trees. Other than that, the counter also helps users of Ecosia see an increased sense of connection among each other and a sense of belonging to a larger tree-planting project. This feeling of community drives users to stick with the site and support reforestation initiatives.



Feature 2: Transparency and Impact Reports:

Target Behavior: Increase user engagement with Ecosia's transparency

Ecosia provides transparency reports and updates on tree planting projects funded by their revenue. Users can access detailed information about where trees are planted, the types of projects supported, and the environmental benefits achieved

Ability - Ecosia's website is designed to make it easy for users to access and understand impact reports. The information provided in the reports is organised clearly and presented in an attractive and understandable style. By providing users a straightforward way to acquire and understand the information without requiring complicated navigation, this design improves user ability. To make the data easier for viewers to understand, the reports may, for instance, contain interactive maps, charts, and graphs.

Motivation - Transparency reports give insight on how users' search behaviours have affected the real world, including the number of trees grown and environmental benefits achieved. Users are encouraged to keep using Ecosia as a sustainable search engine by this information since it makes it clear how their activities are having a beneficial effect. Users may also feel proud of themselves after reading the reports, which might motivate them even more to support Ecosia's purpose. For instance, users could be motivated to keep using Ecosia to have an even bigger impact if they realise that their searches contributed in planting thousands of trees.

WHERE ARE OUR TREES BEING PLANTED?

We plant in 35+ countries with local organizations



Brazil

Planting projects in Brazil protect thousands of endangered plants and animals.

[Learn more](#)



Senegal

Planting trees in Senegal helps restore desertified land to its former fertility.

[Learn more](#)

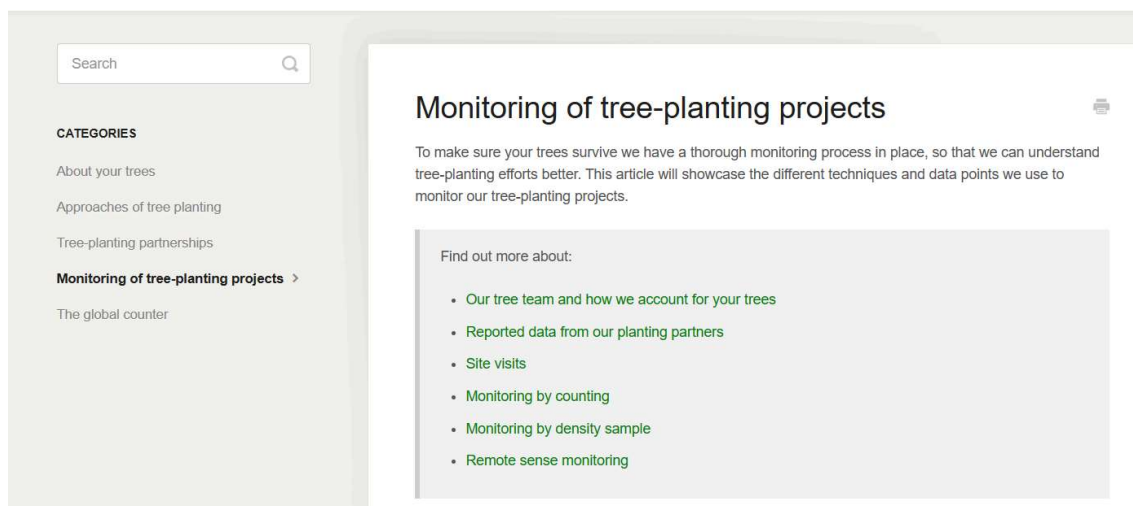


Indonesia

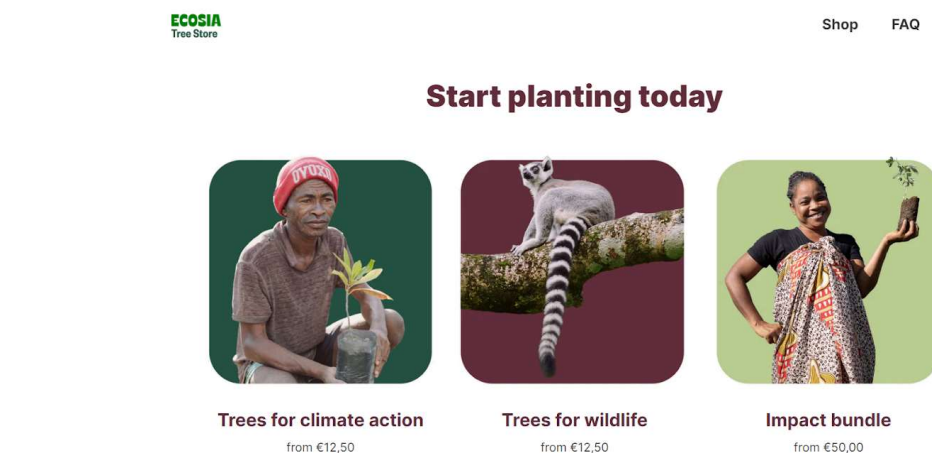
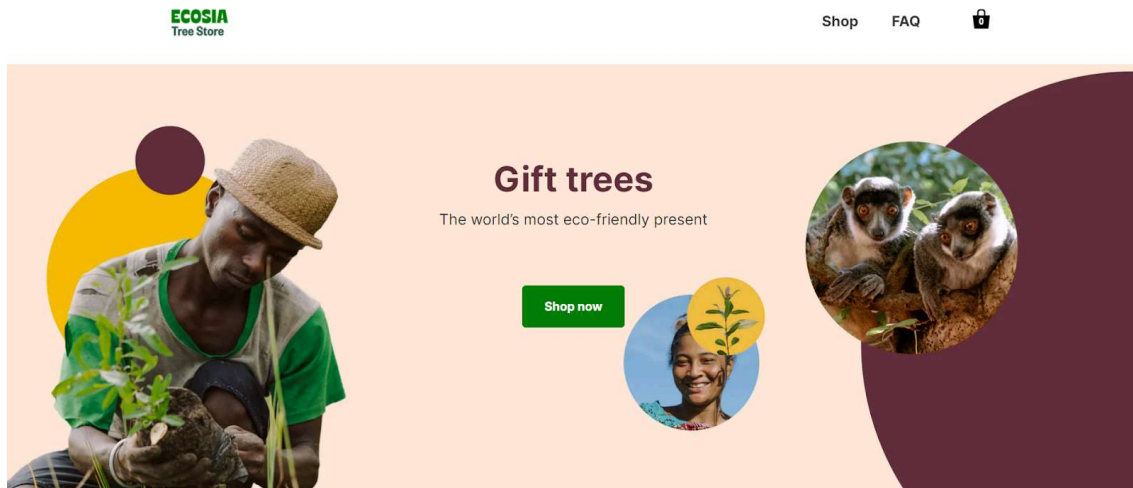
In Indonesia, we help restore forests on former palm oil plantations while creating alternative sources of income.

[Learn more](#)

Trigger - Regular updates and reports about tree planting progress and impact serve as triggers for users to stay engaged and informed about Ecosia's contributions to SDG 15. Users are encouraged to visit Ecosia's website and discover more about the environmental effect of their searches by receiving notifications or emails about fresh reports. The trigger works well to keep users interested and emphasise how their choices continue to have an impact. To entice people to read the entire impact report on the website, for instance, they may receive a monthly newsletter that contains excerpts from the most recent report.



d) One element that may fail to encourage desired behaviour, which is to monitor the success of tree-planting activities on Ecosia, is the absence of rewards for individual contributions to tree planting. According to the FBM, there are a few variables that could prevent this website from achieving long-term growth.



Motivation - Ecosia's current homepage design displays the overall number of trees grown by the organisation as well as the amount allocated to climate action. This serves as a general motivator for visitors to continue using this website by allowing them to understand their individual impact on environmental protection. However, it does not offer individualised encouragement to individual users. This absence of individualised motivation may result in diminished motivation for some users who contribute or take initiatives in order to receive rewards (incentive theory of motivation) or people who are not well-versed in environmental protection. If people do not perceive the instant value in utilising Ecosia, they may change their preferred search engine to another.

Ability - While Ecosia has a basic and intuitive layout, the website's ability can be enhanced by adding more interactive features connected to individual contributions. Ecosia, for example, can include capabilities that allow users to track the location and growing status of trees they have helped plant. However, technical restrictions such

as delayed loading times and device-specific problems can make it difficult for consumers to properly use and interact with the functionality.

Trigger - Ecosia currently lacks appropriate triggers, which are cues that activate behaviour, either internal or external, in order to motivate users to utilise the platform on a consistent basis and contribute to tree planting. The absence of phrases on the platform indicating rewards for individual contributions may induce users to neglect the choice of making a contribution to the tree planting endeavour. As a result, Ecosia's aim is unlikely to be realised.

Improvement - Some designs that we recommend Ecosia implement include a personalised dashboard, an interactive map, and achievement badges. Introduced a personalised dashboard that requires users to create profiles and track their individual contributions. It may include information such as the amount of trees planted by the individual, O₂ emissions, CO₂ emissions offset, how trees purify the air, and how they chill the streets as a natural air conditioner. Furthermore, some elements of an interactive map may include the ability to zoom in on the location of individual trees planted, customise what is visible on the map, and modify how it is visualised (various tree patterns signify different types of trees planted). Finally, reward users with achievement badges or tokens based on their search history. Users can select which badges to display on their homepage to reflect their sense of accomplishment. Ecosia can also provide seamless connectivity with popular social media platforms like Facebook, Instagram, Twitter, and Tik Tok, allowing users to share their achievements. As a result, more netizens will notice this website, which uses social triggers to increase user engagement and exposure.